

Experiment No. 3

Test Case 1

Design a quarter-wave transformer for an input impedance (Z_{in}) of 50 ohms and a load impedance (Z_L) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

PROGRAM:

```
clear; clc;
ZL=100; Zin=50; f=[100,200,300,500]*10^6;
Ro=sqrt(Zin*ZL);
printf("characteristic impedance Ro = %f ohms\n",Ro);
for i=1:4
    L=300/(f(i)*10^-6)/4;
    printf("f=%f MHz    length = %f m\n",f(i)*10^-6,L);
end
```

OUTPUT:

$Z_L = 100$ ohms, $Z_{in} = 50$ ohms $\rightarrow$ Characteristic impedance $R_o = 70.7107$ ohms (Simulation) / 70.7107 ohms (Theoretical)

Frequency (MHz)	Length of $\lambda/4$ transformer (m) — Simulation	Length of $\lambda/4$ transformer (m) — Theoretical
100	0.7500	0.7500
200	0.3750	0.3750
300	0.2500	0.2500
500	0.1500	0.1500

OUTPUT SCREENSHOT:

The screenshot displays the MATLAB R2020b environment. The Command Window shows the execution of the script, outputting the characteristic impedance $R_o = 70.7107$ ohms and the quarter-wave lengths for frequencies 100, 200, 300, and 500 MHz. The Variable Browser shows the values of the variables L , Z_L , Z_{in} , Z_o , f , and λ . The Command History shows the sequence of commands executed.

Command Window Output:

```
characteristic impedance Ro = 70.7107 ohms
f=1e+08 MHz    length = 0.7500 m
f=2e+08 MHz    length = 0.3750 m
f=3e+08 MHz    length = 0.2500 m
f=5e+08 MHz    length = 0.1500 m
```

Variable Browser:

Name	Value	Type	Volatility	Memory
L	0.15	Double	local	236 K
ZL	100	Double	local	236 K
Zin	50	Double	local	236 K
Zo	70.7	Double	local	236 K
f	[100, 200, 300, 500]	Double	local	240 K
i	4	Double	local	236 K
lambda	0.6	Double	local	236 K

Command History:

```
// -- 09/07/2026 11:45:55 -- //
// -- 09/07/2026 11:46:28 -- //
// -- 09/07/2026 11:46:19 -- //
// -- 28/07/2026 08:57:56 -- //
// -- 28/07/2026 09:24:37 -- //
// -- 28/07/2026 10:24:44 -- //
// -- 29/07/2026 13:22:23 -- //
```

Test Case 2

For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms, 75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance ($R_o = \sqrt{Z_{in} \cdot Z_L}$) changes with Zin.

PROGRAM:

```
clear; clc;
ZL=100; f=2*10^9; Zin=[25,50,75,100];
for i=1:4
    Ro=sqrt(Zin(i)*ZL);
    L=300/(f*10^-6)/4;
    printf("Zin=%f Ro=%f Length=%f m\n",Zin(i),Ro,L);
end
```

OUTPUT:

ZL = 100 ohms, f = 2.0 GHz — Variable: Zin

Zin (Ω)	Ro (Ω) — Simulation	Length (m) — Simulation	Ro (Ω) — Theoretical	Length (m) — Theoretical
25	50.0000	0.0375	50.0000	0.0375
50	70.7107	0.0375	70.7107	0.0375
75	86.6025	0.0375	86.6025	0.0375
100	100.0000	0.0375	100.0000	0.0375

OUTPUT SCREENSHOT:

The screenshot displays the MATLAB Scilab 2025.1.0 interface. The main window shows the 'Scilab 2025.1.0 Console' with the following output:

```
QUARTER WAVE TRANSFORMER DESIGN
Frequency = 2.0 GHz
Load Impedance (ZL) = 100 Ohms

Zin(Ohms)  Ro(Ohms)  Length (m)
-----
25          50.0000    0.0375
50          70.7107    0.0375
75          86.6025    0.0375
100         100.0000    0.0375
```

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
L	0.0375	Double	local	216 K
Ro	100	Double	local	216 K
ZL	100	Double	local	216 K
Zin	[25, 50, ...]	Double	local	240 K
f	2e+09	Double	local	216 K
i	4	Double	local	216 K

The Command History shows the following commands:

```
// -- 09/07/2026 11:46:55 -- //
// -- 09/07/2026 11:46:56 -- //
// -- 09/07/2026 11:46:57 -- //
// -- 09/07/2026 08:57:56 -- //
32
33
// -- 28/07/2026 09:24:37 -- //
// -- 28/07/2026 10:54:44 -- //
// -- 29/07/2026 13:22:23 -- //
```